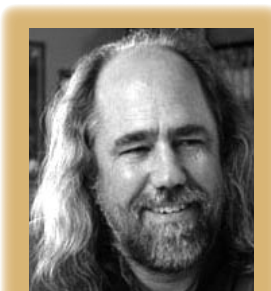


Software Abundance in the Face of Economic Scarcity, Part I

Grady Booch

For a few years now, an image has circulated on the Web of a homeless man holding a sign labeled “Will code HTML for food.”¹ John Dvorak has spoken up as the one behind this photo—it was part of an April Fool’s joke—but the image has taken on a life of its own and is typically accepted as true. While staged, it does put a human face on the matter.



That being said, by no means am I making light of this man’s or anyone’s plight: the human cost of the current global economic malaise is very real and quite devastating. I myself have friends, hard-working people with good hearts and simple intentions, who while not yet on the streets are dangerously close to it. As reported in the *Economist* (4–10 Apr. 2009), the Organization for Economic Cooperation and Development projects that the gross domestic product of its 30 most affluent member nations will drop on average by 4.3 percent in 2009. The World Bank projects the world’s economy to shrink by 1.7 percent this year. At the same time, we sit in a veritable cornucopia of information and technology. According to one report, “A week’s worth of the *New York Times* contains more information than a person was likely to come across in a lifetime in the 18th century.”² It took 38 years for radio to reach a market audience of 50 million, but it took Facebook only two.

Now, I’m neither an economist nor a social scientist, so I can’t speak with any authority about these things. However, I can attend to where software might fit into this picture, and here’s where I land: software-intensive systems are an inescapable and necessary element in helping us operate, innovate, and even thrive in the face of lean economic times.

Let me explain. This is a long story; well, no, actually, it’s a short story that just takes a long time

to tell (Leonard Gershe, *Butterflies Are Free*, 1970). So, I’ll let this theme unfold over two columns.

Scarcity

The bad economic news continues to roll in. The market has lost more than eight trillion dollars in value in the past year. Housing prices continue to fall, unemployment continues to climb, and many economists project that recovery will take a decade or more. The phrase “Great Recession” has gained traction.³

And yet, we’ve been here before.⁴ There have been three periods of great volatility in the market and one serious bear market between 1996 and the present. If you go back further, the Dow Jones Industrial Average has lost more than 40 percent of its value nine times since 1885.⁵ We had the collapse of the Russian economy in 1998, the bursting of the dot-com bubble around 1995, and the Great Depression of 1929. In a more distant generation, there was also

- the Panic of 1893 (triggered by serious rot in the railroad business),
- the Panic of 1857 (triggered by bank failures), and even
- the South Sea Bubble of 1720 (led by speculation in the stock of a company that had taken on much of England’s debt from the War of Spanish Succession).

As Barbara Tuchman notes in *A Distant Mirror* (Ballantine Books, 1987), the 14th century was not a particularly happy time, what with the plague and constant wars all across Europe, and I definitely would not want to have been hanging around Rome in 476. In his book *Collapse* (Viking, 2004), Jared Diamond has similarly detailed the collapse of several societies, some of whom seemingly evap-

orated from the face of the earth.

In the face of economic scarcity, the traditional response is to retreat, shrink, slash (and try not to panic). As General Motors began its freefall, it was reported that “to avoid bankruptcy, the government had said GM must shed debt, cut labor costs, and close plants.”⁶ Even that was too little too late, and GM filed for bankruptcy protection on 1 June.

Abundance

There’s another, more creative reaction to economic scarcity: to attack. While the velocity of money may have greatly decelerated and while credit might still be a scarce resource, human imagination is not similarly constrained. As Clay Shirky has observed,⁷ we have a cognitive surplus: producing all of Wikipedia, in all its forms and languages, took about 100 million thought-hours, and yet in the US alone, watching television consumes about 200 billion thought-hours per year, the equivalent of 2,000 Wikipedias.

There’s certainly a tension between abundance and scarcity. Innovation in times of abundance is relatively easy; even just staying alive in lean times is a struggle. The thing about abundance is that, well, abundance feels a whole lot better than scarcity. However, there’s a dark side. When you are a reasonable distance away from the financial edge, you feel more confident in taking risks. When you are really far away from the edge you get fat and lazy. When you don’t even believe that there is an edge, you get arrogant and reckless.

As Brad Perkins has noted, “A recession exacerbates competition for limited resources.” I agree. But unless this competition results in complete starvation, there are always opportunities on the edge.⁸

Software Abundance

So, here’s the thing about software. It’s perhaps the most fungible and liquid of resources. The supply of software is limited only by human imagination and labor. Software has no mass (although I’d argue that it does have weight), it’s remarkably malleable and wonderfully versatile, and it never wears out. This is why I also argue that software-intensive systems are an inescapable and necessary element in helping us operate, innovate, and even thrive.

Now, there are skeptics in every age. In

his wonderful lecture, “Getting the Network the World Needs,” Lawrence Lessig quotes the composer John Philip Sousa as saying “these talking machines are gong to ruin the artistic development of music in this country.”⁹ More recently, the chairman and CEO of a major entertainment company declared himself to be “a guy who sees nothing good having come from the Internet. Period.”¹⁰ Ouch. While I can see why he might say that from his worldview, I have to respectfully disagree.

Undeniably, software-intensive systems, even in these lean economic times, are the force behind a sea change in communication and connectivity and a force that continues to propel innovation. As Ethan Zuckerman noted in his incredible riff on “The Cute Cat Theory of Digital Activism,”¹¹

The Web was invented so that physicists could share research papers. Web 2.0 was invented so we could share cute pictures of our cats. The tools of Web 2.0, while designed for mundane uses, can be extremely powerful in the hands of digital activists, especially those in environments where free speech is limited.

He goes on to observe that you can use Flickr to post cute pictures of your cat, YouTube for cute videos of your cat, Google Maps to specify where you took those cute pictures and videos, Twitter to post real-time updates on the cute things your cat is doing, and then of course Blogger to wax philosophic about those pictures and videos and tweets. I’d add to his list things such as Facebook, where one can build a global social network of people who admire your cat.

Of course, these technologies are not meant only for banal purposes. Zuckerman observes that Flickr can also be used to reveal subversive photographs that would otherwise be blocked by firewalls. YouTube can be used to post videos of trade union protests or, I’d add, police brutality, governmental stupidity, or commercial crime. Google Maps, he notes, can be used to overlay records of land ownership, sometimes disclosing otherwise nonobvious connections. Twitter can be used to offer real-time updates about events, perhaps to coordinate a flash mob. Blogger can be used as a platform for reporting news, sub-

verting traditional media that may otherwise be at best inadequate or at worst controlled by a suppressive government.

Lessig characterizes this fundamental change as one in which the consumer is now the creator, leading us to what he calls a read/write culture. Such platforms make possible new and unexpected uses of media, such as D.J. Spooky’s remix of D.W. Griffith’s 1915 film *Birth of a Nation*, a most remarkable artistic achievement.¹²

It’s not just about entertainment, however. In the second part of this column (in the Nov./Dec. issue), I’ll examine other places where software abundance can rise above economic scarcity, and then turn to the role of stable, yet flexible architectures that serve as platforms for innovation. 

References

1. J. Dvorak, “Will Code HTML for Food;” www.html4.com/mime/markup/html/will.html.
2. “Did You Know ?” *Global Entrepreneurship Week/USA*, 5 May 2009; <http://entrepreneurshipweekusa.com/?p=436>.
3. “Economix,” *New York Times*, 11 Mar. 2009; <http://economix.blogs.nytimes.com/2009/03/11/great-recession-a-brief-etymology>.
4. “A Visual History of the Stock Market from 1996 to 2007,” *Generation X Finance*, 26 Nov. 2007; <http://genxfinance.com/2007/11/26/a-visual-history-of-the-stock-market-from-1996-2007>.
5. M.J. Lundeen, “Dow Jones Industrials –40% Declines 1885 to 2008,” 12 Oct. 2008; www.gold-eagle.com/editorials_08/lundeen101308.html.
6. “GM Bondholders Balk at Plan; Chapter 11 Now Likely,” *ABC News*, 27 May 2009; http://news.yahoo.com/s/ap/20090527/ap_on_bi_ge/us_gm_bondholders.
7. C. Shirky, “Here Comes Everybody,” 26 Apr. 2008; www.shirky.com/herecomeseverybody/2008/04/looking-for-the-mouse.html.
8. H. Macleod, *Gapingvoid: Cartoons Drawn on the Back of Business Cards*, 5 Sept. 2008; www.gapingvoid.com/edges005.jpg.
9. L. Lessig, “Getting the Network the World Needs,” 24 Mar. 2009; <http://blip.tv/file/1937322>.
10. M. Lynton, “Guardrails for the Internet: Preserving Creativity Online,” *Huffington Post*, 26 May 2009; www.huffingtonpost.com/michael-lynton/guardrails-for-the-internet_b_207459.html.
11. E. Zuckerman, “The Cute Cat Theory of Digital Activism,” 4 Mar. 2008; <http://en.oreilly.com/et2008/public/schedule/detail/1597>.
12. D.J. Spooky’s *Rebirth of a Nation*; www.djspooky.com.

Grady Booch is an IBM Fellow and one of the UML’s original authors. He also developed the Booch method of software development (*Object-Oriented Analysis and Design*, Addison-Wesley, 1993). He’s working on a handbook of architectural patterns, available at www.booch.com/architecture. Contact him at architecture@booch.com.